Dropout-Induced Stochasticity for GRPO Fine-Tuning of Latent-Reasoning Language Models

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April 28, 2026

Abstract

Latent-reasoning language models such as Coconut (Hao et al., 2024) replace discrete chain-of-thought tokens with continuous hidden states fed recurrently into the next reasoning step. Because the latent recurrence is deterministic given the input, naïve policy-gradient methods such as GRPO cannot generate diverse rollouts in latent space, which removes the variance that group-relative advantage estimation relies on. We propose to use a small dropout rate $p \approx 0.1$ as the source of stochasticity during rollout, and to *replay the same dropout mask* during the policy update so that the latent trajectory observed at rollout time is reproduced exactly. We show that, under this construction, dropout noise plays the role of an *exogenous* reparameterization variable, the per-mask GRPO surrogate is an unbiased estimator of the policy gradient of an augmented (mask-marginalized) policy, and replaying the mask provides common-random-numbers variance reduction over the alternative of resampling at update time. We also give an inductive argument that the surrogate gradient is well defined at every latent depth T, and we relate the construction to the MC-dropout Bayesian interpretation of dropout.

1 Introduction and motivation

Group Relative Policy Optimization (GRPO; Shao et al., 2024) has become the standard reinforcement-learning objective for fine-tuning reasoning LLMs on math benchmarks. Its appeal is that the per-prompt advantage is computed purely from a group of K rollouts and a verifier reward, with no learned critic. GRPO requires that the K rollouts *differ*, because the group-normalized advantage,

$$A^{(k)} = \frac{r^{(k)} - \mu_r}{\sigma_r + \delta}, \qquad \mu_r = \frac{1}{K}\sum_{k=1}^{K} r^{(k)}, \qquad \sigma_r^2 = \frac{1}{K}\sum_{k=1}^{K}(r^{(k)} - \mu_r)^2, \tag{1}$$

is identically zero whenever all rollouts agree. In a standard token-level LLM, diversity is supplied automatically by sampling from the next-token distribution.

Coconut-style latent reasoning, by contrast, does not sample. Hidden states $h_t \in \mathbb{R}^d$ are fed back into the model as the embedding of the next "token" for $t = 1, \dots, T$ until a stopping criterion fires; only after the latent phase ends does the model emit discrete answer tokens $y_1, \dots, y_L$. Conditional on the prompt x, the latent phase is a *deterministic* function of the parameters θ. This breaks GRPO: all K rollouts share the same h_T and (for greedy decoding) the same y.

We propose to inject stochasticity by enabling dropout during rollout. This yields per-rollout latent trajectories that depend on a Bernoulli mask $\xi \sim p(\xi)$ which is independent of θ, so GRPO can grade rollouts relative to one another. At update time we re-mount the *same* mask so that the gradient is computed against the trajectory that was actually graded. The rest of this note formalizes the construction and proves the basic correctness and variance properties.

2 Preliminaries

2.1 Coconut recurrence with dropout

Let $f_\theta : \mathbb{R}^d \to \mathbb{R}^d$ denote one Transformer step of the latent-reasoning model with parameters $\theta \in \mathbb{R}^D$. Standard Coconut unrolls

$$h_t \;=\; f_\theta(h_{t-1}), \qquad h_0 \;=\; \mathrm{embed}_\theta(x), \qquad t = 1, \dots, T. \tag{2}$$

With dropout at rate p and keep-probability $q := 1 - p$, define a Bernoulli mask $m_t \in \{0,1\}^D$ with $m_t \sim \mathrm{Bern}(q)^{\otimes D}$ and the rescaled per-step parameters

$$\tilde{\theta}_t(\xi_t) \;:=\; \theta \odot m_t / q, \qquad \mathbb{E}_{\xi_t}[m_t/q] = \mathbf{1}. \tag{3}$$

The dropout-perturbed recurrence is

$$h_t \;=\; f_{\tilde{\theta}_t(\xi_t)}(h_{t-1}), \qquad \xi := (\xi_1, \dots, \xi_T) \sim p(\xi) = \prod_{t=1}^{T} p(\xi_t). \tag{4}$$

Once the latent phase finishes, answer tokens are sampled autoregressively,

$$y_\ell \;\sim\; \pi_\theta(\cdot \mid x, h_T, y_{<\ell}), \qquad \ell = 1, \dots, L, \tag{5}$$

optionally with dropout enabled here as well; we absorb any such sampling randomness into ξ by reparameterizing the categorical samples with Gumbel noise when convenient. A scalar reward $r(y, x) \in [0, 1]$ is provided by an external verifier (e.g. exact-match on GSM8K).

2.2 Augmented policy

The induced distribution over (ξ, y) given x is

$$\Pi_\theta(\xi, y \mid x) \;=\; p(\xi)\, \pi_\theta(y \mid x, \xi), \qquad \pi_\theta(y \mid x, \xi) := \prod_{\ell=1}^{L} \pi_\theta\big(y_\ell \,\big|\, x, h_T(\theta, \xi, x), y_{<\ell}\big). \tag{6}$$

We treat Π_θ as the policy that GRPO optimizes; the dropout mask is part of the action. The marginal policy

$$\bar{\pi}_\theta(y \mid x) \;:=\; \mathbb{E}_{\xi \sim p}\big[\pi_\theta(y \mid x, \xi)\big] \tag{7}$$

is the object whose expected reward we ultimately wish to maximize.

2.3 GRPO objective

For a group of K rollouts $\{(\xi^{(k)}, y^{(k)})\}_{k=1}^{K}$ generated under parameters θ_{old}, with rewards $r^{(k)} = r(y^{(k)}, x)$, define advantages $A^{(k)}$ via (**??**) and the per-token importance ratio

$$\rho_\theta^{(k)}(\xi, y) \;:=\; \frac{\pi_\theta(y \mid x, \xi)}{\pi_{\theta_{\mathrm{old}}}(y \mid x, \xi)}. \tag{8}$$

The clipped GRPO surrogate is

$$\mathcal{L}^{\mathrm{GRPO}}(\theta) \;=\; \mathbb{E}_{x \sim \mathcal{D}} \frac{1}{K} \sum_{k=1}^{K} \min\Big(\rho_\theta^{(k)} A^{(k)},\; \mathrm{clip}(\rho_\theta^{(k)},\, 1 - \varepsilon,\, 1 + \varepsilon)\, A^{(k)}\Big) \;-\; \beta\, \mathrm{KL}\big(\pi_\theta \,\big\|\, \pi_{\mathrm{ref}}\big). \tag{9}$$

3 Method: dropout as exogenous noise with mask replay

The proposed procedure is:

1. For each prompt x, draw K independent dropout masks $\xi^{(1)}, \ldots, \xi^{(K)} \sim p(\xi)$.
2. For each k, run the latent-reasoning recurrence (**??**) with mask $\xi^{(k)}$ *stored on device*, decode answer tokens $y^{(k)}$, compute reward $r^{(k)}$ and advantage $A^{(k)}$ via (**??**).
3. During the policy update, recompute $\pi_\theta(y^{(k)} \mid x, \xi^{(k)})$ with the *same* stored mask $\xi^{(k)}$, so that h_T at update time matches h_T at rollout time exactly when $\theta = \theta_{\text{old}}$. Form the GRPO loss (**??**) and step θ.

The construction rests on three observations.

(i) Independence of ξ from θ. The dropout mask ξ is sampled from a fixed distribution $p(\xi)$ that does not depend on θ. Hence, in (**??**), $\nabla_\theta \log p(\xi) = 0$, and

$$\nabla_\theta \log \Pi_\theta(\xi, y \mid x) \;=\; \nabla_\theta \log \pi_\theta(y \mid x, \xi). \tag{10}$$

The gradient signal therefore comes entirely from the ξ-conditional log likelihood of the answer tokens, which is differentiable through the unrolled latent path.

(ii) Reparameterization view. With ξ fixed, $h_T = h_T(\theta, \xi, x)$ is a deterministic, almost everywhere differentiable function of θ. This is precisely the setting of the reparameterization trick (Kingma & Welling, 2014). Equivalently, sampling ξ first and then computing h_T is a coupling of stochastic processes across different θ values — the common-random-numbers construction (Glasserman & Yao, 1992).

(iii) Mask replay = common random numbers. At update time we evaluate π_θ on the same $\xi^{(k)}$ that was used at rollout time. This couples the rollout-time and update-time trajectories so that the only source of difference between them is the parameter update $\theta - \theta_{\text{old}}$ itself. Without replay, the importance ratio $\rho_\theta(\xi, y)/\rho_\theta(\xi', y)$ would also reflect mask resampling, inflating its variance.

4 Theoretical analysis

4.1 Well-definedness of the surrogate gradient (induction on T)

Assumption 1. For every fixed mask ξ, the map $\theta \mapsto f_{\tilde\theta(\xi)}(h)$ is differentiable in θ for almost every h, and the answer-token log-likelihood $\log \pi_\theta(y \mid x, h_T, y_{<\ell})$ is differentiable in both θ and h_T.

This is satisfied by every standard Transformer block: the dropout rescaling $\theta \odot m/q$ is linear in θ, and the remaining nonlinearities (LayerNorm, GELU, softmax) are almost everywhere differentiable.

Lemma 2 (Well-defined latent gradient at depth T)**.** *Under Assumption* **??***, for every $T \geq 0$ and almost every ξ, the Jacobian $\partial h_T/\partial\theta$ exists and is given by the recursion*

$$\frac{\partial h_t}{\partial \theta} \;=\; \frac{\partial f_{\tilde\theta_t}}{\partial \theta}\bigg|_{h_{t-1}} \cdot \frac{\partial \tilde\theta_t}{\partial \theta} \;+\; \frac{\partial f_{\tilde\theta_t}}{\partial h}\bigg|_{h_{t-1}} \cdot \frac{\partial h_{t-1}}{\partial \theta}, \qquad \frac{\partial h_0}{\partial \theta} \;=\; \frac{\partial\, \mathrm{embed}_\theta(x)}{\partial \theta}, \tag{11}$$

where $\partial\tilde\theta_t/\partial\theta = \mathrm{diag}(m_t)/q$ is the dropout-rescaled identity. Consequently $\nabla_\theta \log \pi_\theta(y \mid x, \xi)$ is well defined.

Proof. By induction on t.

Base case ($t = 0$). $h_0 = \mathrm{embed}_\theta(x)$ is differentiable in θ by assumption, giving the stated initial Jacobian.

Inductive step. Assume $\partial h_{t-1}/\partial\theta$ exists. Write $h_t = f(\tilde{\theta}_t, h_{t-1})$, viewing f as a function of two arguments. By the multivariate chain rule and Assumption **??**,

$$\frac{\partial h_t}{\partial \theta} \;=\; \frac{\partial f}{\partial \tilde{\theta}_t} \cdot \frac{\partial \tilde{\theta}_t}{\partial \theta} \;+\; \frac{\partial f}{\partial h_{t-1}} \cdot \frac{\partial h_{t-1}}{\partial \theta},$$

and $\partial\tilde{\theta}_t/\partial\theta = \mathrm{diag}(m_t)/q$ from (**??**). Both terms exist (the second by the inductive hypothesis), giving (**??**).

Finally, since $\log \pi_\theta(y \mid x, h_T, y_{<\ell})$ is differentiable in both θ and h_T, one more chain-rule application yields $\nabla_\theta \log \pi_\theta(y \mid x, \xi) = \sum_\ell [\partial_\theta \log \pi_\theta]_{\mathrm{direct}} + [\partial_{h_T} \log \pi_\theta] \cdot \partial h_T/\partial\theta$. □

Remark 3. The recursion (**??**) is what `loss.backward()` implements when the dropout masks are stored and replayed: PyTorch's `torch.nn.functional.dropout` treats the rescaled mask as a constant during backprop, which is exactly $\partial\tilde{\theta}_t/\partial\theta = \mathrm{diag}(m_t)/q$.

4.2 Unbiasedness of the dropout-GRPO surrogate

Theorem 4 (Unbiased policy gradient on the augmented policy)**.** *Let* $J(\theta) := \mathbb{E}_{x\sim\mathcal{D}}\, \mathbb{E}_{(\xi,y)\sim\Pi_\theta(\cdot|x)}[r(y,x)]$ *be the expected reward under the augmented policy* (**??**)*. Then*

$$\nabla_\theta \mathcal{L}^{\mathrm{GRPO}}(\theta)\Big|_{\theta=\theta_{\mathrm{old}}} \;=\; \mathbb{E}_{x,\xi,y}[A(\xi,y)\, \nabla_\theta \log \pi_\theta(y \mid x, \xi)]_{\theta_{\mathrm{old}}} \;=\; \nabla_\theta J(\theta)\Big|_{\theta_{\mathrm{old}}}, \tag{12}$$

provided the KL term in (**??**) *is evaluated as* $\mathrm{KL}(\Pi_\theta \| \Pi_{\mathrm{ref}})$ *on the augmented policy.*

Proof. At $\theta = \theta_{\mathrm{old}}$ we have $\rho_{\theta_{\mathrm{old}}} \equiv 1$, so the clip is inactive and the gradient of the min reduces to the gradient of the unclipped ratio:

$$\nabla_\theta \rho_\theta\big|_{\theta_{\mathrm{old}}} \;=\; \nabla_\theta \log \pi_\theta(y \mid x, \xi)\big|_{\theta_{\mathrm{old}}}.$$

Therefore

$$\nabla_\theta \mathcal{L}^{\mathrm{GRPO}}\big|_{\theta_{\mathrm{old}}} \;=\; \mathbb{E}_{x,\xi,y}[A(\xi,y)\, \nabla_\theta \log \pi_\theta(y \mid x, \xi)]_{\theta_{\mathrm{old}}}. \tag{13}$$

By (**??**), $\nabla_\theta \log \pi_\theta(y \mid x, \xi) = \nabla_\theta \log \Pi_\theta(\xi, y \mid x)$, so the right-hand side is the REINFORCE estimator on the augmented policy with advantage A. The group-relative baseline μ_r is independent of any single rollout's score function in expectation (a standard control-variate identity), so it contributes zero bias. Hence the right-hand side equals $\nabla_\theta J(\theta)|_{\theta_{\mathrm{old}}}$. □

Corollary 5 (Bias on the marginal policy)**.** *The estimator is unbiased for* $J(\theta)$*, i.e. for the expected reward of the mask-marginalized policy* (**??**)*, since* $\mathbb{E}_\xi[\pi_\theta(y \mid x, \xi)] = \bar{\pi}_\theta(y \mid x)$ *implies* $\mathbb{E}_{x,\xi,y\sim\Pi_\theta}[r] = \mathbb{E}_{x,y\sim\bar{\pi}_\theta}[r]$*. In other words, optimizing the augmented policy is exactly optimizing expected reward under the dropout ensemble.*

4.3 Variance reduction via mask replay (CRN)

Consider two estimators of $\nabla_\theta J$ at a perturbed parameter $\theta = \theta_{\mathrm{old}} + \Delta$:

$$\hat{g}_{\mathrm{CRN}} := A(\xi, y)\, \nabla_\theta \log \pi_\theta(y \mid x, \xi), \quad \xi \text{ shared with rollout}, \tag{14}$$

$$\hat{g}_{\mathrm{indep}} := A(\xi, y)\, \nabla_\theta \log \pi_\theta(y \mid x, \xi'), \quad \xi' \perp\!\!\!\perp \xi \text{ resampled}. \tag{15}$$

Proposition 6 (CRN variance reduction)**.** *Suppose the trajectory-conditional reward $\bar{r}(\xi) := \mathbb{E}_y[r \mid \xi]$ and the score function $s(\xi, y; \theta) := \nabla_\theta \log \pi_\theta(y \mid x, \xi)$ are positively correlated across ξ, i.e. $\mathrm{Cov}_\xi(\bar{r}(\xi),\, s(\xi, y; \theta)) \succeq 0$ component-wise. Then*

$$\mathrm{Var}[\hat{g}_{\mathrm{CRN}}] \;\leq\; \mathrm{Var}[\hat{g}_{\mathrm{indep}}]. \tag{16}$$

Proof sketch. Both estimators have the same mean (Theorem **??**). Decompose each variance into a within-mask and across-mask component using the law of total variance. The within-mask components are equal. For the across-mask component, the resampling estimator $\hat{g}_{\mathrm{indep}}$ has uncorrelated $\bar{r}(\xi)$ and $s(\xi', y; \theta)$ (since $\xi' \perp \xi$), whereas $\hat{g}_{\mathrm{CRN}}$ has covariance equal to the assumed nonnegative quantity. Subtracting yields the bound. This is the standard CRN argument (Glasserman & Yao, 1992, Thm 3.1) specialized to the score-function estimator. □

Remark 7. The positive-correlation assumption is mild in practice: masks that yield a correct answer ($\bar{r}(\xi)$ large) also yield a score function that pushes θ in directions that increase π_θ on those tokens. Empirical verification can be done by tracking the sign of the diagonal of $\mathrm{Cov}_\xi(\bar{r}, s)$ during training.

4.4 Connection to MC-dropout

Following Gal & Ghahramani (2016), interpret each mask as a posterior sample $\theta_\xi := \theta \odot m(\xi)/q$ from a variational distribution $q_\theta(\theta')$ over parameters. The marginal policy (**??**) is then

$$\bar{\pi}_\theta(y \mid x) \;=\; \mathbb{E}_{\theta' \sim q_\theta}[\pi_{\theta'}(y \mid x)], \tag{17}$$

i.e. a Bayesian-model-average policy. Theorem **??** states that dropout-GRPO performs ensemble-aware policy optimization: each gradient step is the Monte Carlo average of per-ensemble-member policy gradients, with the group-normalized advantage acting as a control variate.

5 Algorithm

Algorithm 1: Dropout-GRPO for latent-reasoning LLMs.

Inputs: base policy $\pi_{\theta_{\mathrm{old}}}$, prompts $\mathcal{D}$, group size K, dropout rate p, latent depth T, clip ε, KL coef β, mini-epochs E.

1. For each prompt $x \sim \mathcal{D}$:

   a. *Rollout* (train mode, dropout on). For $k = 1, \ldots, K$: sample fresh dropout RNG seeds, store seeds for $\xi^{(k)}$, unroll latent recurrence (**??**), decode $y^{(k)}$, record per-token logprobs $\log \pi_{\theta_{\mathrm{old}}}(y^{(k)} \mid x, \xi^{(k)})$, compute $r^{(k)} = \mathrm{verifier}(y^{(k)}, x)$.

   b. Compute advantages $A^{(k)} = (r^{(k)} - \mu_r)/(\sigma_r + \delta)$ as in (**??**).

   c. *Update* (mask replay). For mini-epoch $e = 1, \ldots, E$ and each k: regenerate $\xi^{(k)}$ from stored seeds, recompute $\log \pi_\theta(y^{(k)} \mid x, \xi^{(k)})$, form $\rho^{(k)} = \pi_\theta/\pi_{\theta_{\mathrm{old}}}$, take a gradient step on $\mathcal{L}^{\mathrm{GRPO}}$ from (**??**).

   d. Set $\theta_{\mathrm{old}} \leftarrow \theta$.

6 Practical concerns and open questions

Sensitivity of h_T to ξ. If dropout perturbations to h_T do not change the argmax of the answer-token distribution, all K rollouts agree, advantages collapse to zero, and no gradient is produced. This is the dropout-rate analogue of temperature in token-level RL. A diagnostic to monitor is the within-group standard deviation of $r^{(k)}$ (Eq. (**??**)); if $\sigma_r \to 0$, raise p or fold token-sampling Gumbel noise into ξ.

Importance-ratio explosion. The ratio $\rho_\theta(\xi, y)$ couples parameter updates and dropout-induced logit shifts, and can be large because dropping different units changes logits non-smoothly. The PPO-style clip ε may need to be tightened relative to the token-LM regime, or applied per latent step rather than only per answer token.

Validity of clip under CRN. The clip operator in (**??**) is non-smooth, so Theorem **??** as stated holds only at $\theta = \theta_{\text{old}}$ (initial step). Standard PPO analysis (Schulman et al., 2017, §5) extends the bound to small steps under a Lipschitz condition on ρ_θ, which in turn requires ρ_θ to remain bounded under mask replay; this is plausible but should be verified empirically (track max ratio).

Train/deploy mismatch. Inference typically disables dropout, so the deployed policy is $\pi_\theta(y \mid x, \xi = \mathbf{1})$ rather than the marginal $\bar{\pi}_\theta$. Two options: (a) keep dropout on at deploy time and average over masks, or (b) accept a bias bounded by $\mathrm{TV}(\bar{\pi}_\theta, \pi_\theta(\cdot \mid \mathbf{1})) = \mathcal{O}(p\sqrt{d})$ under standard regularity assumptions on the network's Lipschitz constant in its mask argument.

Memory for stored masks. Storing $\xi^{(k)}$ for K rollouts and T latent steps over a D-parameter model costs $\mathcal{O}(KTD)$ bits. In practice, only the dropout layers' masks need to be stored (not full-parameter masks), and a per-layer RNG seed is sufficient: store the seed, regenerate the mask deterministically at update time. This is the recommended implementation.

7 Summary

- Latent-reasoning models break GRPO because the latent phase is deterministic; dropout supplies the missing stochasticity without changing the model class.
- Dropout noise is exogenous (independent of θ), so it acts as a reparameterization variable; the score function reduces to the ξ-conditional log likelihood (Eq. **??**).
- The latent gradient $\partial h_T / \partial \theta$ is well defined at every depth T (Lemma **??**), justifying backprop through the unrolled latent recurrence.
- GRPO with replayed masks is an unbiased estimator of the policy gradient on the dropout-augmented policy, hence on the marginal policy (Theorem **??**).
- Replaying the rollout-time mask at update time gives a CRN coupling that lower-bounds variance against the resample-at-update alternative (Proposition **??**).
- Practical caveats are: h_T-sensitivity, importance-ratio control, clip-region Lipschitz behavior, and train/deploy mismatch.

References (informal)

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